SURAKSHA

Delivery Partner Insurance

Technical README & Product Documentation

Version 2.0 | Suraksha Insurance Tech Pvt. Ltd. | 2025

One Insurance. Every Delivery Partner. Every Platform.

15 Lakh+	**Rs.29/week**	**60 Seconds**	**8 Platforms**
Eligible Partners	Starting Premium	Max Payout Time	Under One Policy

IRDAI Compliant **RBI PPI Licence** **DPDP Act 2023** **Auto-payout in 60s**

1. Executive Summary

Suraksha is India's first universal parametric micro-insurance product for the gig delivery workforce. It provides automated, real-time income protection for delivery partners working across multiple platforms under a single affordable weekly policy. When weather makes delivery dangerous, Suraksha pays the partner automatically within 60 seconds with no form, no assessor, and no waiting.

Unlike traditional insurance products that require claim submission and weeks of processing, Suraksha uses live environmental data (weather, AQI, visibility) to automatically trigger payouts the moment conditions are confirmed. Partners never need to file anything.

The Core Promise

When weather makes delivery dangerous, Suraksha pays you automatically -- before you even know it triggered.

1.1 The Problem Being Solved

Gap	Impact on Partners
No income protection on bad weather days	Partners lose 50-70% of daily earnings (Rs.800-1,500) on extreme weather events
Traditional insurance does not fit gig workers	Irregular income, no employer, 2-4 week claim cycles, frequent rejections
Multi-platform workers fall through the gaps	Working Swiggy + Zepto means zero income protection from either on the same bad weather day
AQI and heat risk is invisible and ignored	Delhi/Lucknow partners ride in AQI above 200 for 60+ days per year with zero compensation
No real-time trigger product exists in India	Zero insurance products use live environmental data to auto-trigger payouts -- everything is manual

1.2 Market Opportunity

Metric	Value
Total addressable gig delivery workforce in India (2024)	12-15 lakh active partners
Swiggy active delivery partners	3,00,000+
Zomato active delivery partners	3,50,000+
Amazon Flex delivery associates	1,20,000+
Other platforms combined (Zepto, Blinkit, Dunzo, Porter, Delhivery)	3,40,000+
High-impact weather days per city per year	40-60 days (trigger potential)
Average daily earnings at risk per event	Rs.800 to Rs.1,500 per partner

2. Product Overview

Suraksha is a parametric insurance product. Payouts are triggered by pre-defined objective conditions (weather thresholds), not by individual claim assessment. This eliminates the human bottleneck from insurance and enables sub-minute payout speeds at scale.

2.1 Pricing Tiers

Feature	Basic	Plus	Pro	Pro+
Weekly Premium	Rs.29	Rs.59	Rs.99	Rs.149
Heavy Rain Payout	Rs.100/day	Rs.200/day	Rs.350/day	Rs.500/day
AQI / Air Pollution	--	Rs.150/day	Rs.250/day	Rs.400/day
Heat Wave	--	--	Rs.200/day	Rs.350/day
Dense Fog	--	Rs.150/day	Rs.350/day	Rs.400/day
High Wind / Storm	Rs.100/day	Rs.200/day	Rs.350/day	Rs.500/day
Thunderstorm/ Cyclone	Max payout	Max payout	Max payout	Max payout
Accident Cover	Rs.50,000	Rs.1,00,000	Rs.2,00,000	Rs.5,00,000
Max Weekly Payout	Rs.300	Rs.700	Rs.1,400	Rs.2,500
Platforms Linked	Up to 2	Up to 4	Unlimited	Unlimited+
24x7 AI Chatbot	Basic	Standard	Priority	Priority+

All plans renew automatically every Monday via UPI AutoPay. Partners can cancel, upgrade, or downgrade any Monday with zero penalty. All plans include AI chatbot support in Hindi and English, weekly summary dashboard, and live 24x7 zone monitoring.

2.2 Parametric Trigger System

The trigger system polls environmental data every 15 minutes across a 5km x 5km grid. At least 2 of 3 independent data sources must agree before any payout is processed. Partners must be actively logged into at least one linked platform during the trigger window.

Trigger Type	Threshold Condition	Plans Covered	Daily Payout Range
Rain (Heavy)	> 20mm/hr for 30+ min	All tiers	Rs.100-500/day
Air Pollution (AQI)	AQI > 200 for 2+ hrs	Plus, Pro, Pro+	Rs.150-400/day
Heat Wave	Feels-like > 45C for 4+ hrs	Pro, Pro+	Rs.200-350/day
Dense Fog	Visibility < 200m for 2+ hrs	Plus, Pro, Pro+	Rs.150-400/day
High Wind	Wind > 50km/hr for 1+ hr	All tiers	Rs.100-450/day
Thunderstorm/ Cyclone	IMD severe weather warning	All tiers	Max tier + safety alert

2.2.1 Trigger Validation Rules

- **Multi-source consensus:** At least 2 of 3 independent data sources (OpenWeatherMap, IMD Nowcast, CPCB/Weather.com) must confirm the condition before payout fires
- **Zone precision:** Triggers fire at 5km x 5km zone level -- not city-wide averages. Koramangala is unaffected by Whitefield trigger events
- **Duration threshold:** Conditions must persist for the specified minimum duration. A 5-minute rain burst does not qualify as a trigger
- **Active partner check:** Partner must be online on at least one linked platform during the trigger window to prevent gaming
- **Data sources:** OpenWeatherMap API, IMD Nowcast, CPCB AQI stations, Weather.com commercial API, satellite weather feeds

3. How It Works -- End to End

The entire journey from partner sign-up to payout is designed to require zero human intervention on Suraksha's side and minimal friction on the partner's side.

1. **Step 1 -- Link All Platforms:** Partner opens Suraksha inside their Swiggy or Zomato partner app and links all delivery platforms in a single account. All platforms are covered simultaneously under one weekly policy from that point forward.
2. **Step 2 -- Quick eKYC in 2 Minutes:** Aadhaar-based eKYC via DigiLocker auto-fills partner details. Partners already KYC-verified on Swiggy or Zomato skip this step entirely. No documents, no branch visits required.
3. **Step 3 -- Choose Weekly Plan:** Partner selects Basic (Rs.29), Plus (Rs.59), Pro (Rs.99), or Pro+ (Rs.149) per week. Premium auto-deducts every Monday via UPI AutoPay mandate. Cancel, pause, or upgrade any Monday with zero lock-in.
4. **Step 4 -- 24x7 Live Zone Monitoring:** Suraksha polls real-time weather and AQI data every 15 minutes for the partner's active 5km x 5km delivery zone. Monitoring runs continuously across all linked platforms simultaneously. Partners never need to check anything manually.
5. **Step 5 -- AI Trigger Evaluation:** The Trigger Evaluation Engine verifies that conditions meet plan thresholds, cross-validates with 2 of 3 independent data sources, and checks partner activity status. The AI decides and acts -- no human in the loop.
6. **Step 6 -- Payout in 60 Seconds:** Payout lands in the partner's Swiggy, Zomato, or Suraksha wallet within 60 seconds of trigger confirmation. A push notification explains exactly what triggered and how much was credited. No form. No follow-up.
7. **Step 7 -- Weekly Summary Dashboard:** Every week partners receive a summary: payouts received, trigger events fired, days covered, and a 7-day weather forecast for their zone. Suraksha auto-renews every Monday unless cancelled.

4. Weather Alert System

The Weather Alert System is a real-time monitoring layer that tracks environmental conditions across all major Indian cities. It displays live data to partners through the Suraksha app weather dashboard. When conditions approach or cross trigger thresholds, partners receive proactive alerts and safety advisories.

4.1 Live City Monitoring -- Sample Data

City	Temp	Feels	Rain	AQI	Wind	Status	Trigger Status
Mumbai	31C	38C	28mm/hr	82	22km/hr	Rain Alert	**Rain Trigger ACTIVE**
Delhi	38C	44C	0mm/hr	245	12km/hr	AQI Danger	**AQI Trigger ACTIVE**
Chennai	35C	41C	4mm/hr	98	18km/hr	Clear	No active trigger
Hyderabad	42C	47C	0mm/hr	112	9km/hr	Heat Alert	**Heat Wave ACTIVE**
Bengaluru	27C	29C	2mm/hr	68	15km/hr	Normal	No active trigger
Kolkata	33C	40C	14mm/hr	145	24km/hr	Wind Alert	**Wind Trigger ACTIVE**
Pune	29C	33C	6mm/hr	75	11km/hr	Light Rain	No active trigger
Ahmedabad	43C	48C	0mm/hr	168	7km/hr	Extreme Heat	**Heat Wave ACTIVE**

4.2 Alert Threshold Reference

Condition	Trigger Threshold
Heavy Rainfall	Rainfall rate above 20mm per hour, sustained for 30 or more minutes
Air Quality Index (AQI)	AQI above 200 (Very Poor on CPCB scale), sustained for 2 or more hours
Heat Index (Feels-Like)	Feels-like temperature above 45 degrees C, sustained for 4 or more hours

Dense Fog (Visibility)	Visibility below 200 metres, sustained for 2 or more hours
High Wind Speed	Wind speed above 50 km per hour, sustained for 1 or more hour
Thunderstorm / Cyclone	IMD Severe or Cyclone Warning issued for the partner's zone

4.3 Safety Advisories Sent to Partners

Condition	Safety Advisory
Heavy Rain	Wear waterproof gear. Reduce speed by 30%. Avoid flooded roads. Wait it out if rainfall is extreme.
Poor Air Quality	Wear N95 mask. Limit outdoor exposure. Hydrate frequently. Avoid high-traffic corridors.
Extreme Heat	Carry 2L water minimum. Rest every 90 minutes. Wear light clothing. Avoid peak afternoon hours.
High Wind	Reduce bag load. Avoid bridges and open roads. Keep both hands on handlebars at all times.
Thunderstorm / Cyclone	STOP deliveries immediately. Move to covered shelter. Do not resume until IMD warning is lifted.

5. AI & Machine Learning Engine

Seven interdependent AI and ML systems underpin Suraksha's operations. These systems work in concert to ensure payouts are fast, accurate, fraud-resistant, and financially sustainable across 15 lakh partners.

1. Trigger Evaluation Engine

Rule engine + Random Forest classifier

Validates zone conditions against plan thresholds using 3 independent data sources. 2-of-3 consensus required before payout fires. Payout decision delivered in under 60 seconds with no human in the loop. Target: below 0.5% false payout rate.

2. City-Level Threshold Calibration

LSTM time-series model

Uses 3 years of historical weather data, delivery volume, and incident records to calibrate per-city trigger thresholds. Bengaluru's monsoon threshold differs meaningfully from Delhi's. Model re-calibrates every quarter.

3. Multi-Platform Fraud Detection

Isolation Forest + GPS anomaly detection

Monitors every payout event across all linked platforms in real time. Detects GPS spoofing, fake activity signals, and coordinated gaming attempts. Operates across platform boundaries simultaneously.

4. Risk-Based Dynamic Pricing

LSTM + gradient boosting regression

Forecasts zone-level environmental risk 7 days ahead. High-risk weeks such as cyclone season or extreme AQI periods trigger adjusted premium calculations to maintain actuarial viability through monsoon season.

5. Multi-Platform Partner Intelligence

Graph neural network model

Builds a unified partner activity profile across all linked apps. Correctly attributes zone, activity, and payout eligibility when a partner is simultaneously active on multiple platforms at the same time.

6. AI Chatbot -- Hindi & English 24x7

Claude API (Anthropic) + LangChain + Pinecone RAG

Answers partner queries in Hindi and English instantly. Context-aware: a query like 'Why did I not get paid yesterday?' returns an accurate, specific answer using the partner's actual trigger log, zone data, and policy details.

7. Actuarial Risk Modelling

Monte Carlo simulation + XGBoost regression

Projects expected claim rates, payout volumes, and loss ratios by zone, platform, and season. Feeds pricing decisions and reinsurance structuring. Re-runs every quarter with fully refreshed data.

6. Partner Platform Integration

Suraksha covers partners across India's entire gig delivery ecosystem under a single weekly policy -- regardless of which platform's bag they carry. One policy. All platforms. All triggers.

Platform	Category	Active Partners	Notes
Swiggy	Food & Grocery Delivery	3,00,000+	Largest platform
Zomato	Food Delivery	3,50,000+	Largest partner count
Zepto	10-min Grocery	30,000+	Fast-growing
Amazon Flex	E-commerce	1,20,000+	Nationwide
Blinkit	Quick Commerce	40,000+	Rapid expansion
Dunzo	Hyperlocal	20,000+	Metro cities
Porter	Logistics	50,000+	B2B + B2C
Delhivery	E-commerce Logistics	2,00,000+	Pan-India

6.1 B2B API Integration

Delivery platforms integrate Suraksha via a fully-documented REST API and SDK in 3-5 engineering days. Suraksha handles KYC, premium collection, trigger evaluation, payouts, compliance, and reporting end-to-end.

- **API Integration (3-5 engineering days):** Full REST API with Node.js, Python, and Java SDKs. Sandbox environment and dedicated integration engineer provided at no extra cost.
- **Encrypted Zone Data Sync:** Platforms share anonymised zone-level data for trigger zone mapping. AES-256 encrypted, used exclusively for trigger evaluation -- never for partner profiling or marketing.
- **Optional Premium Co-Funding:** Platforms can co-fund partner premiums as a welfare benefit -- e.g., covering Basic tier for all active partners as a recruitment differentiator.
- **Wallet Integration for Payouts:** Direct Suraksha payouts into the platform's existing partner wallet. Partners see payouts with platform branding; Suraksha handles all insurance accounting behind the scenes.
- **Real-Time B2B Analytics Dashboard:** Zone-level trigger events, payout volumes, coverage rates, and 7-day weather risk forecasts useful for surge pricing, fleet routing, and demand forecasting.

7. Payment & Onboarding System

The Suraksha payment and onboarding flow is designed for zero friction. A partner can go from first launch to fully protected in under 5 minutes on a 2G mobile connection.

7.1 Partner Onboarding -- 3 Steps

Step	Details
Step 1: Confirm Plan	Partner selects their weekly plan. Plan details, covered features, and example payout scenarios are shown clearly before any commitment.
Step 2: Link Platforms	Partner checks all delivery platforms they work on. Suraksha verifies account linkage via secure OAuth. Platform credentials are never stored by Suraksha.
Step 3: Payment	Partner sets up UPI AutoPay mandate via GPay, PhonePe, or Paytm, or opts for Swiggy or Zomato wallet auto-deduct, or Rupay/ Visa debit card. Weekly premium auto-deducts every Monday. Cancel anytime with zero penalty.

7.2 Supported Payment Methods

Payment Method	Details
UPI AutoPay (Recommended)	Mandate via GPay, PhonePe, Paytm, or any BHIM UPI app. One-time approval then auto weekly debit every Monday.
Swiggy / Zomato Wallet	Direct deduction from existing platform wallet. No new payment instrument required. Partners with positive wallet balance activate instantly.
Rupay / Visa / Mastercard Debit	Recurring debit mandate. Supported for all nationalised and private banks in India.

7.3 Payout Flow -- Step by Step

1. Trigger confirmed by AI Trigger Evaluation Engine (2 of 3 data sources agree, partner active on linked platform).

2. Payout amount calculated based on partner plan tier and the specific trigger type that fired.
3. Amount credited to partner's linked wallet (Swiggy / Zomato / Suraksha) within 60 seconds.
4. Push notification sent to partner with trigger type, amount credited, and zone affected.
5. Payout logged in partner's weekly dashboard for full transparency and tax records.
6. Weekly summary notified every Monday with all trigger events, payout breakdown, and 7-day forecast.

8. Regulatory & Compliance Framework

Suraksha is built on a full regulatory foundation from day one. Every layer -- insurance distribution, payments, data handling, and security -- is compliant with applicable Indian regulations.

Regulatory Area	Framework & Approach
IRDAI Insurance Distribution	Embedded insurance distributor via Digit Insurance / ICICI Lombard. Delivery platforms registered as IRDAI Corporate Agents -- no additional licensing burden for platforms.
RBI PPI Wallet Licence	Payout wallet under a Prepaid Payment Instrument licence via NBFC partner, regulated by RBI. Auto-debit under RBI UPI AutoPay e-mandate framework.
DPDP Act 2023	Explicit informed consent at onboarding. Data retention capped at 24 months. Right to erasure, portability, and grievance redressal fully supported in-app.
Zone-Level Data Only	All trigger data is zone-level aggregate -- never personal. GPS data for anti-fraud is processed on-device, encrypted AES-256, never shared with any delivery platform.
Platform Data Isolation	Data from each platform siloed in separate encrypted namespaces. Swiggy data is architecturally isolated from Zomato data -- enforced at infrastructure level.
End-to-End Security	All partner data encrypted at rest (AES-256) and in transit (TLS 1.3). API credentials via AWS Secrets Manager. Annual third-party security audit mandated.

9. React Website -- SurakshaApp.jsx

The Suraksha website is built as a fully self-contained single-file React JSX component (SurakshaApp.jsx) using only React hooks and inline JavaScript style objects. No external CSS frameworks are required.

9.1 Component Architecture

Component / Function	Purpose
SurakshaApp (default export)	Root app component with page routing (home / weather / payment) and all global state management.
Chakra	SVG Ashoka Chakra logo with 24 spokes -- pure SVG, configurable size, color, and rotation speed.
Fade	Scroll-reveal wrapper using IntersectionObserver -- triggers fade-and-slide animation on viewport entry.
Num	Animated number counter -- counts from 0 to target value when scrolled into view.
Radar	Canvas-based live radar animation with rotating sweep, zone circles, and moving partner dots (pure HTML5 Canvas).
WeatherPage	Full live weather page: 8-city data table, city selector, 6-metric grid, safety tips.
PaymentPage	3-step payment flow: Plan confirm, Platform linking, UPI/Wallet/Card payment, Success screen.

9.2 Page Routing

Page (useState value)	Content
home	Hero, Marquee, Stats, Problem, How It Works + Radar, Triggers, Platforms, Pricing, AI Engine, Compliance, CTA, Footer
weather	Alert banner, city selector, current conditions card, 6-metric grid, 8-city data table, 4 safety tip cards
payment	3-step checkout: Plan Confirm, Platform Linking (checkbox grid), Payment (UPI/Wallet/Card), Success with policy ID

9.3 Full-Width Fix -- Critical Setup

Required CSS -- add to index.css or main.jsx

Without this, the website shows black side bars on localhost:5173 (Vite default layout contains #root at limited width)

```
/* index.css -- Required for full-width layout */
html, body, #root {
  width: 100%;
  max-width: 100%;
  margin: 0;
  padding: 0;
  overflow-x: hidden;
}
```

9.4 Dependencies & Setup

Dependency	Notes
React 18+	Uses useState, useEffect, useRef hooks only
Vite or Create React App	Any modern bundler -- no special config required for SurakshaApp.jsx
Google Fonts (Playfair Display + Nunito Sans)	Loaded via @import in injected CSS string -- no npm package needed
CSS Framework	None required -- all styles are inline JavaScript objects
React Router	Not needed -- routing is pure useState between three views
Canvas / Chart library	Not needed -- Radar animation uses pure HTML5 Canvas via useRef

10. Design System

Suraksha uses a premium insurance brand aesthetic -- deep navy, warm gold, and cream -- inspired by traditional Indian policy documents. The design language signals trust, authority, and care for the partner.

10.1 Color Palette

Token	Hex / Usage
Navy (Primary)	#0D1B3E -- Hero backgrounds, navigation bar, footer, section accents, headings
Gold (Brand)	#C8962C -- Primary CTA buttons, accent borders, eyebrow labels, trigger highlights
Gold Light	#E8B84B -- Ashoka Chakra logo, hero italic text, featured plan accents, AI section
Cream (Background)	#F7F3EC -- Alternating section background throughout all pages
White	#FFFFFF -- Card backgrounds, table rows, alternating section backgrounds
Text Gray	#7A8BB0 -- Body copy, muted labels, secondary information throughout

10.2 Typography

Usage	Font / Weight
Display headings and brand name	Playfair Display -- serif, weight 700 to 800
Italic emphasis (hero, subtitles)	Playfair Display Italic -- adds warmth and humanity to the brand
Body text, labels, navigation links	Nunito Sans -- weight 300, 400, 600, or 700 depending on context
Eyebrow labels above headings	Nunito Sans 700, 0.6rem, 2.5px letter-spacing, uppercase, gold color
Code and policy IDs	Courier New monospace

10.3 Ashoka Chakra Logo

The Suraksha logo is a custom SVG React component called Chakra -- not an image file. It renders the authentic Ashoka Chakra with 24 equally-spaced spokes, outer ring, inner ring, and centre hub. It rotates continuously via a CSS keyframe animation.

- **Props:** size (pixels), color (hex string), speed (CSS animation-duration string e.g. '8s')
- **Navbar:** 28px gold, rotation speed 8s
- **Hero background:** 480px white, 7% opacity, speed 24s -- decorative background element
- **CTA section background:** 580px white, 5% opacity, speed 28s -- subtle rotating brand element
- **Footer:** 20px gold, speed 8s -- matches navbar for consistency

11. Frequently Asked Questions

Q: What is parametric insurance and how is it different from traditional insurance?

A: Parametric insurance pays out automatically when a pre-defined condition is confirmed by objective data -- no claim submission, no assessor visit, no approval process required. Traditional insurance requires the partner to submit a claim, provide evidence of loss, wait for an assessor, and then accept or appeal a decision. Suraksha's parametric model eliminates all of that. Payouts fire in 60 seconds.

Q: What if I work on Swiggy and Zomato simultaneously?

A: One Suraksha policy covers all linked platforms simultaneously. If you are active on both Swiggy and Zomato during a trigger event, you receive a single payout from Suraksha -- not two separate payouts. The Multi-Platform Partner Intelligence AI model correctly identifies and attributes cross-platform activity to avoid double-counting.

Q: What if weather is bad but I do not get a payout?

A: This happens in three scenarios: (1) The condition did not reach the trigger threshold -- for example, 18mm per hour of rain when the trigger requires 20mm per hour. (2) Only 1 of 3 data sources confirmed the condition, so 2-of-3 consensus was not reached. (3) You were not active on any linked platform during the trigger window. The AI Chatbot can explain exactly why a specific event did not trigger for your zone and plan.

Q: Can I cancel my plan?

A: Yes -- you can cancel any Monday in-app with zero penalty. Your coverage continues until the end of the current week. There is no lock-in, no cancellation fee, and no notice period required.

Q: How secure is my platform login data?

A: Suraksha does not store your platform login credentials at any point. We use OAuth-based platform linking -- you authorise Suraksha to receive anonymised zone-level data only. Your delivery history, earnings, ratings, and personal data remain entirely within each platform's systems.

Q: Is Suraksha registered with IRDAI?

A: Yes. Suraksha operates as an embedded insurance distributor under IRDAI's corporate agent framework, partnered with Digit Insurance and ICICI Lombard as the licensed insurers. The underlying insurance policies are regulated IRDAI products, not informal or unregistered financial products.

12. Contact & Legal

Suraksha Insurance Tech Pvt. Ltd.

Registered in India | IRDAI Embedded Distributor | RBI PPI Licence | DPDP Act 2023 Compliant

Detail	Information
Company	Suraksha Insurance Tech Pvt. Ltd.
Product	Suraksha Parametric Delivery Partner Insurance
Version	2.0 (React JSX Edition -- SurakshaApp.jsx)
Insurance Partners	Digit Insurance / ICICI Lombard (IRDAI Licensed Insurers)
Payment Infrastructure	Razorpay + UPI AutoPay (RBI Regulated)
Data Protection	DPDP Act 2023 -- Consent, Retention, Portability, Erasure
Source Code File	SurakshaApp.jsx -- Single-file React component, Version 2.0
Document Version	README v2.0, March 2025

This document is confidential and intended for internal use, investor review, and platform partner evaluation only.

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